Problem Statement : The AI Energy Efficiency Optimization Agent for Mechanical Systems

The Challenge

Mechanical systems such as HVAC units, compressors, turbines, and industrial machines consume large amounts of energy. Inefficient operation increases operational costs and environmental impact. Organizations often lack advanced tools to analyze energy consumption patterns and identify opportunities for optimization.

The Objective

Build an AI-powered Energy Efficiency Optimization Agent that analyzes mechanical system data and recommends improvements to reduce energy consumption.

1. **Energy Data Retrieval (RAG) :–** Collect energy consumption data and operational parameters.
2. **Efficiency Analysis :–** Identify inefficiencies in machine operations.
   Optimization Recommendations :– Suggest operational adjustments or system improvements.
3. **Energy Forecasting :–** Predict future energy consumption trends.
4. **Multi-Agent System**
   - Energy Monitoring Agent – Collects energy consumption data.
   - Efficiency Analysis Agent – Detects inefficient operations.
   - Optimization Agent – Recommends energy-saving strategies.
   - Forecasting Agent – Predicts future energy usage.
5. **Visualization Dashboard**
   Display energy consumption trends, efficiency indicators, and optimization recommendations.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models